Notes: Ferreira, Matos, and Pires (JF, 2018)

Asset Management within Commercial Banking Groups: International Evidence

Contents

1 Big picture

- **Question.** Do mutual funds run inside commercial banking groups serve fund investors, or do they also support the commercial bank's lending business?
- **Institutional setting.** Around 40% of mutual funds worldwide are run by asset management companies connected to commercial banking groups, and the phenomenon is especially important outside the United States.
- **Three hypotheses.** The paper contrasts a conflict-of-interest hypothesis, an information-advantage hypothesis, and a null hypothesis in which Chinese walls keep lending and asset management independent.
- **Main performance result.** Commercial bank-affiliated equity funds underperform unaffiliated funds by about 92 basis points per year, or roughly 23 basis points per quarter.
- **Conflict mechanism.** Underperformance is stronger when affiliated funds overweight stocks of firms that borrow from the parent bank, especially the top lending clients.
- **Causal evidence.** Divestitures of asset management divisions by commercial banking groups reduce client-stock holdings and are associated with improved fund performance.
- **Client-stock evidence.** Client stocks bought by high-bias affiliated funds underperform, which is inconsistent with the bank transmitting superior private information to the fund.
- **Bank-benefit evidence.** Bank-affiliated funds' client-stock holdings increase the probability that the parent bank is chosen for future syndicated lending business.
- **Governance evidence.** Funds affiliated with lending banks vote less against management on compensation proposals, consistent with supporting borrower relationships.
- **Career-concern channel.** Fund managers who support the bank's lending business appear less likely to be dismissed after poor performance.
- **Investor discipline.** Flows respond to past performance, but affiliated funds' conflicts may be partly sustained because parent-bank distribution networks and investor inattention weaken discipline.
- **Economic takeaway.** Combining commercial banking and asset management creates cross-division spillovers: fund investors may bear costs when fund managers support the bank's lending relationships.

2 Data and measurement

2.1 Fund sample and affiliation classification

- The sample covers actively managed domestic equity mutual funds domiciled in 28 countries from 2000 to 2010.
- Domestic funds are the focus because commercial banks are most likely to have lending relationships with domestic firms.
- A fund is **commercial bank-affiliated** if the fund management company is majority-owned by a commercial parent bank or belongs to a group that owns a commercial bank.
- Other fund families are classified as unaffiliated, investment bank-affiliated, or insurance-affiliated.
- The sample contains 4,981 domestic equity funds with $3.57 trillion in TNA as of December 2010.

Interpretation: The key institutional fact is that commercial bank affiliation is common internationally. This creates a large sample in which conflicts between lending and asset management can be tested.

2.2 Fund performance

The main performance measure is the quarterly Carhart four-factor alpha:

$$R_{i,t} - R_{f,t} = \alpha_i + \beta_{MKT} MKT_t + \beta_{SMB} SMB_t + \beta_{HML} HML_t + \beta_{MOM} MOM_t + \varepsilon_{i,t}. \tag{1}$$

The paper also uses benchmark-adjusted returns, gross four-factor alphas, buy-and-hold benchmark-adjusted returns, and information ratios.

Interpretation: A negative coefficient for commercial bank-affiliated funds means that the funds underperform after standard risk and style exposures are removed.

2.3 Client stocks and conflict proxies

A **client stock** is a stock issued by a firm that borrows from the parent bank of a fund's management company.

- %TNA Invested in Client Stocks: fraction of fund assets invested in the bank's lending clients.
- Bias in Client Stocks: overweighting of lending-client stocks relative to comparable passive funds.
- Top 10 Client Stocks: the ten largest lending clients of the parent bank.
- Loans/TNA and Corporate Loans/TNA: measures of the importance of lending activity relative to asset management activity within the banking group.
- Interest Income/Fees: a proxy for how important lending revenues are relative to fee income.

Interpretation: If affiliated managers support the bank's lending business, funds with more exposure to client stocks should perform worse, especially when the parent bank's lending business is large.

2.4 Other bank-benefit and agency variables

- Syndicated loan data identify whether a bank is chosen as a lead arranger after its affiliated funds hold a borrower's stock.
- Proxy voting data identify whether affiliated funds vote against management on executive compensation proposals.
- Manager turnover data identify whether fund managers are punished for poor performance or protected when they help the banking group.
- Flow data identify whether investors discipline underperforming affiliated funds.

Interpretation: These variables let the paper move beyond fund performance and ask who benefits from the conflict: the parent bank, the borrower, or the fund manager.

3 Models and empirical specifications

3.1 Baseline fund-performance regression

The main regression is:

$$Performance_{i,t} = \alpha + \beta CommercialBankAffiliated_{i,t} + \gamma' Controls_{i,t-1} + FE + \varepsilon_{i,t}. \tag{2}$$

Controls include TNA, family TNA, age, expenses, load, flows, number of countries of sale, team management, and past performance. Fixed effects vary by specification.

Interpretation: The coefficient β measures the performance gap between commercial bank-affiliated and unaffiliated funds after controlling for fund characteristics and fixed effects.

3.2 Testing the lending-conflict channel

The paper then interacts affiliation with conflict proxies:

$$Performance_{i,t} = \alpha + \beta CommercialBankAffiliated_{i,t} + \delta HighBiasFund_{i,t} + \gamma' Controls_{i,t-1} + FE + \varepsilon_{i,t}. \tag{3}$$

Similar specifications use high allocation to client stocks and top-10 client-stock measures.

Interpretation: A negative δ means that underperformance is stronger for affiliated funds that hold or overweight the parent bank's lending clients.

3.3 Divestiture difference-in-differences

For divestitures of asset management companies by commercial banking groups, the paper estimates changes around the event:

$$Y_{i,t} = \alpha + \beta After_t + \theta Treated_i + \delta(Treated_i \times After_t) + FE + \varepsilon_{i,t}. \tag{4}$$

The outcomes are client-stock holdings and average fund performance.

Interpretation: If commercial bank ownership creates the conflict, then client-stock holdings should fall and fund performance should improve after the asset manager is divested from the banking group.

3.4 Calendar-time client-stock portfolio tests

Each quarter, client and nonclient stocks are separated into buy and sell portfolios. The paper estimates four-factor alphas on buy-minus-sell portfolio returns.

Interpretation: If affiliated managers have superior information about lending clients, client stocks they buy should outperform client stocks they sell. If they are supporting clients, the opposite can occur.

3.5 Future lending business

The paper estimates logit regressions of the probability that a bank is chosen as lead arranger:

$$Pr(LeadArranger_{b,j,t} = 1) = \Lambda(\alpha + \beta ClientStockHolding_{b,j,t-1} + \gamma' X_{b,j,t-1}). \tag{5}$$

Interpretation: A positive β means that stock support by affiliated funds helps the banking group obtain future lending business.

3.6 Voting dissent, manager turnover, and fund flows

The paper estimates regressions of voting dissent, probit regressions of fund manager turnover, and flow-performance sensitivity regressions:

$$Dissent_{j,t} = \alpha + \beta LenderAffiliatedOwnership_{j,t} + \gamma' X_{j,t} + \varepsilon_{j,t}, \tag{6}$$

$$Turnover_{m,t} = \Lambda(\alpha + \beta ConflictProxy_{i,t} + \gamma Performance_{i,t} + \delta' X_{i,t}), \tag{7}$$

$$Flow_{i,t} = \alpha + \beta Performance_{i,t-1} + \theta Affiliated_i \times Performance_{i,t-1} + \gamma' X_{i,t-1} + \varepsilon_{i,t}. \tag{8}$$

Interpretation: These tests ask whether client support helps borrowers, protects managers, and weakens the usual investor discipline imposed by flows.

4 Identification and empirical design

4.1 What identifies the baseline result

- The baseline compares affiliated and unaffiliated funds over time and across countries.
- Controls absorb fund size, family size, age, expenses, flows, loads, distribution reach, team management, and past performance.
- Fixed effects absorb quarter, country, and sometimes fund-level heterogeneity.
- Standard errors are clustered at the ultimate-owner level.

Interpretation: The baseline performance gap documents a robust association. The paper then uses mechanism tests and divestitures to strengthen causal interpretation.

4.2 Why client-stock tests matter

- If affiliated funds underperform because banks have poor fund-management skill, client-stock exposure should not matter much.
- If affiliated funds underperform because they support lending clients, high-bias and high-allocation funds should underperform more.
- The paper finds exactly this pattern.

Interpretation: The client-stock evidence ties underperformance to the bank's lending relationships rather than to generic poor fund management.

4.3 Why divestitures are important

- Divestitures create a before-after change in commercial-bank ownership.
- The paper matches divested funds to similar nondivested funds.
- Client-stock holdings fall after divestiture.
- Performance improves after divestiture.

Interpretation: Divestitures are the closest design to a causal shock because the fund management company moves away from bank ownership.

4.4 Why alternative explanations are weak

- The information-advantage story predicts that client stocks bought by affiliated funds should outperform; the calendar-time tests do not support this.
- The Chinese-wall/null hypothesis predicts little connection between lending relationships and fund holdings; the client-stock and lending tests reject this.
- Country-level institutions matter, but the conflict appears in both U.S. and non-U.S. samples, with stronger effects outside the United States.

Interpretation: The empirical patterns point to conflicts of interest rather than private information or complete separation of bank divisions.

4.5 Limits of the design

- Fund affiliation is not randomly assigned.
- Bank lending relationships and fund holdings may be jointly determined.
- Some data, especially holdings and voting data, are limited to subsets of countries and time periods.
- Divestitures help but are not a randomized experiment.

Interpretation: The paper's strength is the consistency of evidence across performance, holdings, divestitures, lending, voting, turnover, and flows.

5 Tables and figures

The visuals below are directly cropped from the paper and arranged compactly.

5.1 Table I and Figure 1: market importance of bank-affiliated funds

Read:

- Table I shows that the domestic equity fund sample contains 4,981 funds with $3.57 trillion in assets as of December 2010.
- Commercial bank-affiliated funds are 32.2% of domestic funds and 18.1% of domestic-fund TNA.
- Outside the United States, bank-affiliated funds are 40.3% of funds and 39.8% of TNA.
- Figure 1 shows that bank-affiliated funds remain a large part of the global fund industry, although their market share declines over time.

Economic message: Bank-affiliated asset management is not a niche phenomenon. It is a large international organizational form, especially outside the United States.

Sample of Commercial Bank-Affiliated Funds

This table presents number of funds, total net assets (TNAs), number of ultimate owners, percentage of commercial bank-affiliated funds, and number of parent (commercial) banks as of December 2010 for the sample of open-end actively managed domestic equity mutual funds and for the sample of domestic and international equity mutual funds (bottom of the table).

	Domestic Equity Funds			Commercial Bank-Affiliated Funds		
	Number of Funds	TNA ($ billion)	Number of Ultimate Owners	Number of Funds (%)	TNA (%)	Number of Parent Banks (%)
Australia	98	32.6	28	27.6	16.5	14.3
Austria	13	1.4	11	61.5	81.0	54.5
Belgium	23	1.7	8	73.9	78.6	50.0
Brazil	48	42.0	17	79.2	78.4	58.8
Canada	366	194.6	66	28.4	44.5	21.2
China	69	76.0	35	11.6	8.0	8.6
Denmark	22	3.1	15	54.5	70.0	46.7
Finland	28	5.5	14	71.4	89.8	50.0
France	180	42.2	48	48.9	57.8	27.1
Germany	47	34.8	20	51.1	71.7	45.0
India	242	37.4	31	18.6	17.7	25.8
Israel	37	0.8	15	2.7	1.8	6.7
Italy	30	4.5	15	60.0	55.0	60.0
Japan	515	36.6	43	45.6	36.8	30.2
Malaysia	91	6.4	20	62.6	92.3	45.0
Netherlands	12	4.3	7	66.7	69.9	57.1
Norway	58	15.7	15	58.6	60.2	46.7
Poland	29	5.8	15	58.6	71.0	53.3
Portugal	19	0.5	11	84.2	72.4	81.8
Singapore	13	1.6	10	61.5	28.6	50.0
South Africa	109	21.8	27	38.5	42.3	14.8
Spain	63	2.3	31	65.1	72.4	58.1
Sweden	101	63.2	20	71.3	77.1	40.0
Switzerland	77	20.7	31	55.8	52.1	32.3
Taiwan	147	10.2	31	43.5	26.8	35.5
Thailand	118	5.3	16	62.7	86.0	56.3
United Kingdom	406	215.3	90	17.7	18.0	14.4
United States	2,020	2,683.2	365	20.3	10.9	11.0
Total	4,981	3,569.7	831	32.2	18.1	18.2
Total (ex-U.S.)	2,961	886.5	513	40.3	39.8	25.7
	Domestic and International Equity Funds			Commercial Bank-Affiliated Funds		
Total	10,644	5,842.4	987	40.2	19.9	17.0
Total (ex-U.S.)	7,798	1,897.4	690	47.7	41.2	22.2

(2009), we estimate four-factor alphas using regional factors based on a fund's investment region in the case of domestic, foreign country, and regional funds. We use world factors in the case of global funds.[13]

Table I: sample of commercial bank-affiliated funds

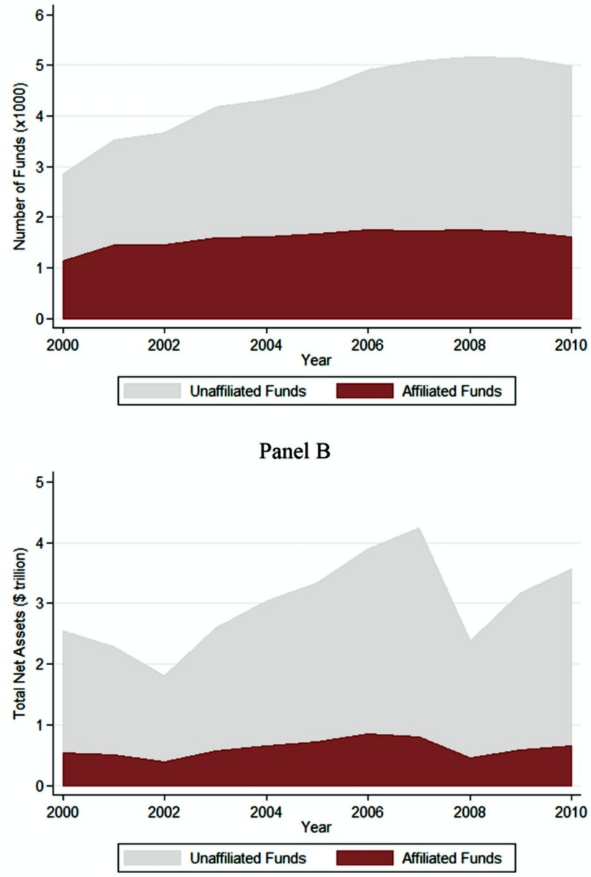

Figure 1: market share of bank-affiliated funds

5.2 Table II: summary statistics

Read:

- Panel A summarizes the full fund sample and conflict proxies.
- Panel B compares unaffiliated and commercial bank-affiliated funds.
- Bank-affiliated funds are smaller on average and have lower benchmark-adjusted returns, lower buy-and-hold benchmark-adjusted returns, and lower flows.
- Panel C shows that bank-affiliated funds invest an average of 14.69% of TNA in client stocks.

Economic message: The summary statistics show substantial client-stock exposure among bank-affiliated funds, which makes the conflict channel economically plausible.

Summary Statistics

Panels A and C present mean, median, standard deviation, 1st percentile, 99th percentile, and number of observations for each variable. Panel B presents mean and number of observations for the samples of unaffiliated funds and commercial bank-affiliated funds, and the corresponding mean difference *p*-value. The sample consists of actively managed domestic equity mutual funds over the 2000 to 2010 period.

Panel A: Fund Characteristics

	Mean	Median	*SD*	1st Percentile	99th Percentile	Number of Observations
Commercial Bank-Affiliated	0.34	0.00	0.47	0.00	1.00	127,880
Publicly Traded Parent	0.64	1.00	0.48	0.00	1.00	127,880
Insurance-Affiliated	0.15	0.00	0.36	0.00	1.00	127,880
Investment Bank-Affiliated	0.22	0.00	0.42	0.00	1.00	127,880
Loans/TNA	36.22	0.00	428.03	0.00	548.92	126,782
Corporate Loans/TNA	26.53	0.00	253.28	0.00	445.74	126,673
Interest Income/Fees	106.56	0.00	792.31	0.00	1,677.93	110,641
%TNA Invested in Client Stocks	5.01	0.00	12.71	0.00	60.16	127,880
%TNA Invested in Top 10 Client Stocks	0.55	0.00	2.40	0.00	12.37	127,880
Bias in Client Stocks	2.01	0.00	7.25	−6.41	39.06	127,238
Bias in Top 10 Client Stocks	0.07	0.00	1.25	−3.14	3.97	127,238
Four-Factor Alpha (%)	0.25	−0.18	5.88	−15.34	19.05	127,880
Benchmark-Adjusted Return (%)	0.06	−0.09	4.18	−12.28	13.61	125,988
Gross-Four-Factor Alpha (%)	0.51	0.09	5.43	−13.73	18.45	116,554
Buy-and-Hold Benchmark Adj.-Return (%)	0.45	0.28	4.12	−12.36	14.78	123,174
Information Ratio	−0.038	−0.057	1.152	−2.825	2.852	127,880
TNA ($ million)	909	158	3,980	2	12,522	127,880
Family TNA ($ million)	35,581	5,501	104,401	15	588,055	127,880
Age (years)	12.46	9.25	11.16	2.33	59.25	127,880
Total Expense Ratio (%)	1.44	1.38	0.57	0.31	3.50	127,880
Total Load (%)	2.42	2.00	2.40	0.00	10.84	127,880
Flow (%)	0.61	−1.45	15.45	−33.70	69.92	127,880
Number of Countries of Sale	1.16	1.00	0.84	1.00	4.00	127,880
Team-Managed	0.61	1.00	0.49	0.00	1.00	127,880

(*Continued*)

Ratio, *Total Load*, *Flow*, *Number of Countries of Sale*, *Team-Managed*). Table A.I in the Appendix provides variable definitions.

Panel B of Table II reports the sample means of the variables separately

Table II, Panel A

Table II—*Continued*

Panel B: Unaffiliated and Commercial Bank-Affiliated Fund Characteristics

	Unaffiliated Funds		Commercial Bank-Affiliated Funds		
	Mean	Number of Observations	Mean	Number of Observations	Difference *p*-Value
Publicly Traded Parent	0.49	84,227	0.92	43,653	0.00
Insurance-Affiliated	0.21	84,227	0.04	43,653	0.00
Investment Bank-Affiliated	0.08	84,227	0.50	43,653	0.00
Four-Factor Alpha (%)	0.26	84,227	0.22	43,653	0.26
Benchmark-Adjusted Return (%)	0.11	83,189	−0.04	42,799	0.00
Gross-Four-Factor Alpha (%)	0.53	78,536	0.48	38,018	0.19
Buy-and-Hold Benchmark-Adj. Return (%)	0.49	81,481	0.38	41,693	0.00
Information Ratio	−0.037	84,227	−0.040	43,653	0.74
TNA ($ million)	1,122	84,227	499	43,653	0.00
Family TNA ($ million)	47,024	84,227	13,501	43,653	0.00
Age (years)	12.54	84,227	12.30	43,653	0.00
Total Expense Ratio (%)	1.44	84,227	1.45	43,653	0.04
Total Load (%)	2.52	84,227	2.24	43,653	0.00
Flow (%)	1.02	84,227	−0.17	43,653	0.00
Number of Countries of Sale	1.16	84,227	1.16	43,653	0.31
Team-Managed	0.59	84,227	0.65	43,653	0.00

Panel C: Commercial Bank-Affiliated Fund Characteristics

	Mean	Median	*SD*	1st Percentile	99th Percentile	Number of Observations
Loans/TNA	107.90	22.75	733.56	0.17	1,148.47	42,555
Corporate Loans/TNA	79.18	10.24	432.77	0.10	977.45	42,446
Interest Income/Fees	446.36	120.81	1,574.14	2.18	6,307.21	26,414
%TNA Invested in Client Stocks	14.69	6.61	18.21	0.00	69.28	43,653
%TNA Invested in Top 10 Client Stocks	1.60	0.00	3.90	0.00	18.49	43,653
Bias in Client Stocks	5.89	1.51	11.46	−12.69	51.55	43,400
Bias in Top 10 Client Stocks	0.22	0.00	2.13	−6.94	7.26	43,400

Table II, Panels B–C

5.3 Table III: performance of commercial bank-affiliated funds

Read:

- In Panel A, the baseline coefficient on Commercial Bank-Affiliated is −0.231% per quarter with a t-statistic of −3.92.
- With fund fixed effects, the coefficient is −0.382% per quarter.
- Adding lending-business variables weakens the affiliation dummy, suggesting that lending-business scale explains much of the performance gap.
- Panel B shows underperformance using alternative performance measures.

Economic message: Bank-affiliated funds underperform, and the link to the scale of the parent bank's lending business points toward conflicts of interest.

Performance of Commercial Bank-Affiliated Funds

This table presents estimates of ordinary least squares (OLS) regressions of fund risk-adjusted performance. Panel A presents results in which the dependent variable is the alpha from the Carhart four-factor model in each quarter. Panel B presents results using alternative measures of risk-adjusted performance. *Commercial Bank-Affiliated* is a dummy variable that takes a value of 1 if the ultimate owner of the fund's management company is a commercial banking group, and 0 otherwise. All control variables are lagged by one period. Variable definitions are provided in Table A.I in the Appendix. The sample consists of actively managed domestic equity mutual funds over the 2000 to 2010 period. Robust t-statistics adjusted for clustering at the ultimate owner level are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Four-Factor Alpha

	(1)	(2)	(3)	(4)	(5)
Commercial Bank-Affiliated	-0.231^{***}	-0.382^{**}	-0.093	-0.121	0.126
	(−3.92)	(−2.35)	(−1.00)	(−1.35)	(0.63)
log (1+Loans/TNA)			-0.050^{**}		
			(−2.14)		
log (1+Corporate Loans/TNA)				-0.051^{**}	
				(−1.98)	
log (1+Interest Income/Fees)					-0.074^{**}
					(−1.99)
Publicly Traded Parent	−0.002	−0.010	−0.006	−0.010	−0.004
	(−0.03)	(−0.05)	(−0.10)	(−0.17)	(−0.07)
Insurance-Affiliated	−0.055	−0.138	−0.062	−0.055	−0.057
	(−0.77)	(−0.52)	(−0.93)	(−0.81)	(−0.83)
Investment Bank-Affiliated	0.106^{*}	0.172	0.103^{*}	0.106^{*}	0.146^{*}
	(1.84)	(0.95)	(1.81)	(1.83)	(1.88)
log (TNA)	-0.052^{***}	-0.617^{***}	-0.054^{***}	-0.054^{***}	-0.045^{***}
	(−4.82)	(−15.57)	(−4.94)	(−4.90)	(−3.82)
log (Family TNA)	0.041^{***}	−0.097	0.040^{***}	0.040^{***}	0.040^{***}
	(3.65)	(−1.24)	(3.47)	(3.51)	(3.15)
log (1+Age)	−0.030	-0.323^{*}	−0.026	−0.025	−0.020
	(−1.09)	(−1.71)	(−0.93)	(−0.91)	(−0.69)
Total Expense Ratio	−0.035	−0.073	−0.035	−0.031	−0.010
	(−0.70)	(−0.47)	(−0.69)	(−0.62)	(−0.18)
Total Load	-0.022^{*}	−0.021	-0.024^{**}	-0.025^{**}	-0.041^{***}
	(−1.95)	(−0.49)	(−2.13)	(−2.14)	(−2.75)
Flow	0.007^{***}	0.005^{***}	0.007^{***}	0.007^{***}	0.007^{***}
	(5.35)	(3.66)	(5.36)	(5.38)	(5.07)
Number of Countries of Sale	−0.002		−0.004	−0.004	0.002
	(−0.12)		(−0.19)	(−0.20)	(0.10)
Team-Managed	-0.105^{***}		-0.107^{***}	-0.107^{***}	-0.088^{**}
	(−2.65)		(−2.71)	(−2.71)	(−2.02)
Past Performance	0.026^{***}	-0.017^{**}	0.026^{***}	0.026^{***}	0.027^{***}
	(3.78)	(−2.44)	(3.76)	(3.75)	(3.74)
Quarter Fixed Effects	Yes	Yes	Yes	Yes	Yes
Country Fixed Effects	Yes	No	Yes	Yes	Yes
Fund Fixed Effects	No	Yes	No	No	No
Number of Observations	127,880	127,880	126,782	126,673	110,641
R^2	0.145	0.192	0.146	0.146	0.131

Table III, Panel A: four-factor alpha

Table III—*Continued*

Panel B: Alternative Measures of Performance

	Benchmark-Adjusted Return (1)	Gross-Four-Factor Alpha (2)	Buy-and-Hold Benchmark-Adj. Return (3)	Information Ratio (4)
Commercial Bank-Affiliated	-0.198^{***}	-0.219^{***}	-0.167^{***}	-0.048^{***}
	(−3.75)	(−3.93)	(−3.33)	(−3.79)
Controls	Yes	Yes	Yes	Yes
Quarter Fixed Effects	Yes	Yes	Yes	Yes
Country Fixed Effects	Yes	Yes	Yes	Yes
Fund Fixed Effects	No	No	No	No
Number of Observations	125,920	116,266	120,198	127,880
R^2	0.034	0.174	0.052	0.089

Table III, Panel B: alternative performance measures

5.4 Table IV: cross-country differences

Read:

- Underperformance is stronger outside the United States: −0.332% per quarter for non-U.S. funds versus −0.165% for U.S. funds.
- It is also stronger in civil-law, bank-based, and high banking-concentration countries.
- The performance gap remains negative across all institutional splits.

Economic message: Conflicts are stronger where bank relationships and bank-centered financial systems are more important and where fund-investor protection may be weaker.

Performance of Commercial Bank-Affiliated Funds: Cross-Country Differences

This table presents estimates of ordinary least squares (OLS) regressions of fund risk-adjusted performance. The dependent variable is the alpha from the Carhart four-factor model in each quarter. *Commercial Bank-Affiliated* is a dummy variable that takes a value of 1 if the ultimate owner of the fund's management company is a commercial banking group, and 0 otherwise. In columns (1) and (2), the non-U.S. and U.S. fund groups consist of those funds domiciled outside of the United States and domiciled in the United States. In columns (3) and (4), the civil and common-law fund groups consist of those funds domiciled in civil-law and common-law countries as defined in La Porta et al. (1998). In columns (5) and (6), the bank- and market-based fund groups consist of those funds domiciled in bank- and market-based countries as defined in Demirgüç-Kunt and Levine (2001). In columns (7) and (8), the high and low bank concentration groups consist of those funds domiciled in countries that are above and below the 75th percentile of the distribution of the market share of the top five banks. In columns (9) and (10), the high- and low-fund management company concentration groups consist of those funds domiciled in countries that are above and below the 75th percentile of the distribution of the market share of the top five fund management companies. In columns (11) and (12), the low and high approvals fund groups consist of those funds domiciled in countries that have one and more than one regulatory approval and disclosure requirements in the fund industry as defined in Khorana, Servaes, and Tufano (2005). The regressions include the same control variables (coefficients not shown) as in Table III. All control variables are lagged by one period. Variable definitions are provided in Table A.I in the Appendix. The sample consists of actively managed domestic equity mutual funds over the 2000 to 2010 period. Robust t-statistics adjusted for clustering at the ultimate owner level are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	Country of Domicile		Legal Origin	
	Non-U.S. Funds (1)	U.S. Funds (2)	Civil Law (3)	Common Law (4)
Commercial Bank-Affiliated	−0.332***	−0.165**	−0.322***	−0.185***
	(−3.49)	(−2.55)	(−2.69)	(−2.83)
Controls	Yes	Yes	Yes	Yes
Quarter Fixed Effects	Yes	Yes	Yes	Yes
Country Fixed Effects	Yes	No	Yes	Yes
Number of Observations	50,864	77,016	24,723	103,157
R^2	0.088	0.246	0.147	0.167

	Financial System		Banking Industry	
	Bank Based (5)	Market Based (6)	High Concentration (7)	Low Concentration (8)
Commercial Bank-Affiliated	−0.307**	−0.197***	−0.405***	−0.199***
	(−2.12)	(−3.18)	(−3.60)	(−3.13)
Controls	Yes	Yes	Yes	Yes
Quarter Fixed Effects	Yes	Yes	Yes	Yes
Country Fixed Effects	Yes	No	Yes	Yes
Number of Observations	22,250	105,630	31,821	96,059
R^2	0.136	0.182	0.117	0.191

Table IV, first part

Table IV—*Continued*

	Mutual Fund Industry		Approvals	
	High Concentration (9)	Low Concentration (10)	Low (11)	High (12)
Commercial Bank-Affiliated	−0.325**	−0.168***	−0.309**	−0.226***
	(−2.56)	(−2.80)	(−2.42)	(−3.53)
Controls	Yes	Yes	Yes	Yes
Quarter Fixed Effects	Yes	Yes	Yes	Yes
Country Fixed Effects	Yes	No	Yes	Yes
Number of Observations	32,094	95,786	28,234	99,646
R^2	0.150	0.199	0.095	0.185

the legacy effect of the Glass-Steagal Act, and hence fund investors' rights are better protected (Khorana, Servaes, and Tufano (2005, 2009)). In columns (1)

Table IV, continued

5.5 Table V: performance and allocation to client stocks

Read:

- The affiliation coefficient remains negative across specifications.
- High-bias affiliated funds underperform by an additional −0.120% per quarter.
- High-bias funds in top-10 client stocks underperform by an additional −0.182%.
- High-allocation funds and high-allocation funds in top-10 clients also underperform more.

Economic message: The funds most exposed to the bank's lending clients perform worst, which supports the conflict-of-interest explanation.

Performance of Commercial Bank-Affiliated Funds and Portfolio Allocation to Client Stocks

This table presents estimates of ordinary least squares (OLS) regressions of fund risk-adjusted performance. The dependent variable is the alpha from the Carhart four-factor model in each quarter. *Commercial Bank-Affiliated* is a dummy variable that takes a value of 1 if the ultimate owner of the fund's management company is a commercial banking group, and 0 otherwise. *High Bias Fund* is a dummy variable that takes a value of 1 if an affiliated fund *Bias in Client Stocks* is above the median in a given country and quarter, and 0 otherwise. *Bias in Client Stocks* is the portfolio bias in stocks of firms that borrow from the fund's parent bank versus the average weight of comparable passive funds. *High Allocation Fund* is a dummy variable that takes a value of 1 if an affiliated fund *%TNA Invested in Client Stocks* is above the median in a given country and quarter, and 0 otherwise. *%TNA Invested in Client Stocks* is the percentage of TNA invested in stocks of firms that borrow from the fund's parent bank. *High Bias Fund in Top 10 Client Stocks* and *High Allocation Fund in Top 10 Client Stocks* are dummy variables similarly defined for the top 10 borrowers of the fund's parent bank. All of these variables are set to zero if the fund is unaffiliated. The regressions include the same control variables (coefficients not shown) as in Table III. All control variables are lagged by one period. Variable definitions are provided in Table A.I in the Appendix. The sample consists of actively managed domestic equity mutual funds over the 2000 to 2010 period. Robust t-statistics adjusted for clustering at the ultimate owner level are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	All Funds				Non-U.S. Funds	U.S. Funds
	(1)	(2)	(3)	(4)	(5)	(6)
Commercial Bank-Affiliated	-0.201^{***}	-0.210^{***}	-0.175^{***}	-0.170^{***}	-0.264^{**}	-0.182^{**}
	(−3.17)	(−3.50)	(−2.70)	(−2.72)	(−2.55)	(−2.53)
High Bias Fund	-0.120^{*}				-0.198^{*}	−0.005
	(−1.65)				(−1.87)	(−0.05)
High Bias Fund in Top 10 Client Stocks		-0.182^{**}				
		(−2.38)				
High Allocation Fund			-0.160^{**}			
			(−2.12)			
High Allocation Fund in Top 10 Client Stocks				-0.258^{***}		
				(−2.98)		
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Quarter Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Country Fixed Effects	Yes	Yes	Yes	Yes	Yes	No
Number of Observations	127,238	127,238	127,880	127,880	50,810	76,428
R^2	0.15	0.15	0.15	0.15	0.088	0.247

D. Robustness Checks

Table IA.V in the Internet Appendix presents additional robustness checks

Table V: performance and allocation to client stocks

5.6 Table VI and Figure 2: divestitures of asset management companies

Read:

- Table VI compares divested fund companies to matched controls before and after divestiture from commercial banking groups.
- After divestiture, funds reduce client-stock holdings and performance improves.
- Figure 2 shows the event-time pattern: client-stock holdings fall after divestiture while average performance rises.

Economic message: Divestitures provide the strongest causal evidence. When the banking-group link is broken, the conflict-related portfolio distortion declines.

Table VI

Divestitures of Fund Management Companies by Commercial Banking Groups

able presents estimates of difference-in-differences regressions of a fund's stock portfolio holdings and risk-adjusted performance (four-fact around the three quarters before and the three quarters after the divestiture of a fund management company by a commercial banking grou A shows tests of equality of pretreatment means and medians of treated and control groups. Panel B shows estimates of difference-in-differenc sions of divestitures during the 2000 to 2010 period (columns (1) and (2)), the 2007 to 2009 global financial crisis (columns (3) and (4)), and t o 2010 period but restricting the sample to funds without fund manager turnover in the event window (columns (5) and (6)). Treated funds a funds sold by a commercial bank to a stand-alone management company. A matched control fund is selected for each treated fund. The cont the nearest neighbor (Mahalanobis distance) from the same quarter, country of domicile, and investment objective (Lipper global classificatic ith the closest *TNA*, *Family TNA*, and *Average Performance* (average fund's four-factor alpha in the previous four quarters). *After* is a dumr le that takes a value of 1 in the announcement quarter of a fund divestiture and thereafter. *%TNA Invested in Client Stocks* is the percenta invested in client stocks (i.e., firms that borrow from the fund's parent bank). Variable definitions are provided in Table A.I in the Append imple consists of actively managed domestic equity mutual funds. Robust t-statistics adjusted for clustering at the deal level are reported heses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Panel A: Matched Sample

	Mean			Median		
	Treated	Control	t-Test (p-Value)	Treated	Control	Pearson χ^2 (p-Valu
	911.9	752.6	0.41	251.6	193.3	0.33
TNA	32,940	22,567	0.00	21,489	9,183	0.11
e Performance	0.13	0.06	0.80	0.19	0.34	0.90

(*Continue*

Table VI, Panels A–B

Table VI—*Continued*

Panel B: Difference-in-Differences

	2000 to 2010		2007 to 2009 Global Financial Crisis		Sample without Fund Manager Turnover	
	%TNA Invested in Client Stocks (1)	Average Performance (2)	%TNA Invested in Client Stocks (3)	Average Performance (4)	%TNA Invested in Client Stocks (5)	Average Performanc (6)
d	11.323^{**}	-0.086^{**}	4.444^{***}	−0.019	13.976^{**}	−0.063
	(2.58)	(−2.20)	(5.11)	(−0.32)	(2.71)	(−0.96)
	−1.310	-0.402^{*}	1.996	0.308	0.659	-0.478^{*}
	(−0.41)	(−1.76)	(0.19)	(0.34)	(0.16)	(−1.97)
d × After	-2.371^{***}	0.412^{***}	-3.018^{***}	0.353^{*}	-2.704^{***}	0.384^{**}
	(−4.75)	(4.30)	(−3.88)	(1.87)	(−4.67)	(2.92)
er Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
er of Observations	1,584	1,577	420	420	1,140	1,136
er of Treated Funds	132	132	35	35	95	95
er of Deals	22	22	7	7	15	15
er of Banks	19	19	7	7	12	12
	0.175	0.135	0.041	0.186	0.271	0.157

Table VI, Panel B continued

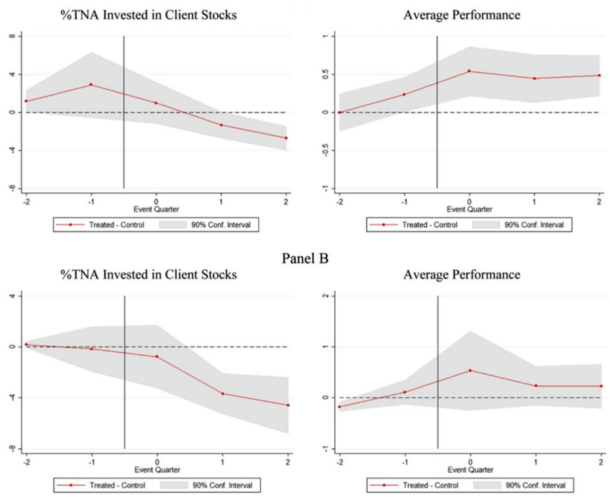

Figure 2. Funds' client stock holdings and performance around divestitures. This figure shows commercial bank-affiliated funds' portfolio holdings of client stocks and performance around divestitures of fund management companies in the 2000 to 2010 (Panel A) and 2007 to 2009 global financial crisis (Panel B) periods. *%TNA Invested in Client Stocks* is the percentage of

Figure 2: client-stock holdings and performance around divestitures

5.7 Table VII: calendar-time portfolio return tests

Read:

- The table compares risk-adjusted monthly returns of client stocks bought and sold by bank-affiliated funds.
- For high-bias funds, client stocks bought underperform client stocks sold.
- The buy-minus-sell alpha for client stocks of high-bias funds is about -0.226% per month.
- The client-minus-nonclient difference is negative for high-bias funds.

Economic message: Client-stock holdings are not explained by superior information. The stock-level evidence is more consistent with client support.

Calendar-Time Portfolio Returns on Buys Minus Sells of Client and Nonclient Stocks

This table presents risk-adjusted monthly portfolio returns of the client stocks a fund buys and sells, defined as the portfolio of client stocks (i.e., firms that borrow from the fund's parent bank) held by bank-affiliated funds that had an increase or decrease in the number of shares held in the previous quarter, respectively. Portfolio returns of the nonclient stocks a fund buys and sells are defined similarly. Every quarter in the 2000 to 2010 period, each fund's portfolio holdings are split into a client portfolio and a nonclient portfolio. These two portfolios are further subdivided into a buy portfolio and a sell portfolio. We calculate the average portfolio return across funds in each month weighted by TNA, and then the return of the portfolio of stocks bought minus sold in each month. Returns are risk-adjusted using the Carhart four-factor model with global factors. The high and low bias fund groups consist of those funds that are above and below the median of the *Bias in Client Stocks* variable in a given country and quarter. The sample consists of actively managed domestic equity mutual funds that are affiliated with commercial banking groups over the 2000 to 2010 period. Robust t-statistics are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	All Bank-Affiliated Funds (1)	High Bias Funds (2)	Low Bias Funds (3)
Client Stocks	−0.115	−0.226*	0.169
	(−0.99)	(−1.90)	(0.68)
Nonclient Stocks	0.033	0.044	0.021
	(0.60)	(0.65)	(0.32)
Client − Nonclient Stocks	−0.148	−0.269*	0.148
	(−1.23)	(−1.90)	(0.61)

underperformance for both client and nonclient stocks would indicate that affiliated fund managers have less skill than unaffiliated fund managers.[23]

We compute the value-weighted monthly portfolio return in quarter t of client stocks in which a fund increased its holdings (in terms of number of

Table VII: calendar-time returns on client and nonclient stocks

5.8 Table VIII: future lending business

Read:

- Client Stock Holding has positive and significant coefficients in the lead-arranger choice regressions.
- Client Stock Holding $> 1\%$ is also positive and significant.
- The implied change in probability of being chosen as lead lender is economically meaningful, especially when there is a past lending relationship.

Economic message: Fund support of client stocks appears to benefit the commercial bank by increasing future lending business.

Table VIII

Probability of Getting Future Lending Business and Client Stock Holdings

This table presents estimates of logit regressions of whether the existence of a bank-firm(i, j) link through bank-affiliated funds' portfolio holdings prior to the loan affects the probability that firm (borrower) j chooses bank i as a lead arranger in the syndicated loan market. For each facility, there is a choice set of 20 potential lead arrangers (top 20 ranked by U.S. dollar volume of syndicated loans in each country). The dependent variable is a dummy variable that takes the value of 1 if bank i acted as a lead arranger, and 0 otherwise. *Client Stock Holding* is a dummy variable that takes the value of 1 if the funds affiliated with bank i hold stock of the firm at the end of the previous year, and 0 otherwise. *Client Stock Holding>1%* is a dummy that takes the value of 1 if the funds affiliated with bank i hold at least 1% of the firm's shares outstanding at the end of the previous year, and 0 otherwise. *Bank Market Share* is the fraction of bank i of the U.S. dollar volume of syndicated loans in each country. *Lending Relationship* is a dummy that takes the value of 1 if firm j chose bank i as a lead arranger in a loan over the past three years. Firm-level controls include stock market capitalization (log), book-to-market ratio, leverage, tangibility, stock volatility, and stock return (coefficients not shown). All control variables are lagged by one period. Variable definitions are provided in Table A.I in the Appendix. The sample consists of syndicated loans by publicly listed borrowers over the 2000 to 2010 period. Robust t-statistics adjusted for clustering at the firm and bank levels are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	(1)	(2)	(3)	(4)
Client Stock Holding	0.269***		0.193***	
	(5.72)		(3.03)	
Client Stock Holding > 1%		0.339***		0.324***
		(3.56)		(3.75)
Bank Market Share	13.266***	13.522***	13.586***	13.824***
	(22.67)	(23.50)	(16.68)	(15.96)
Lending Relationship	1.911***	1.946***	1.748***	1.750***
	(27.33)	(29.07)	(24.61)	(24.79)
log(Bank Assets)			0.119	0.108
			(1.27)	(1.11)
Bank Return on Assets			0.095	0.105
			(1.14)	(1.31)
Year Fixed Effects	Yes	Yes	Yes	Yes
Loan Purpose Fixed Effects	Yes	Yes	Yes	Yes
Bank Fixed Effects	No	No	Yes	Yes
Firm Controls	No	No	Yes	Yes
Firm Industry Fixed Effects	Yes	Yes	Yes	Yes
Firm Country Fixed Effects	Yes	Yes	Yes	Yes
Number of Observations	499,143	499,143	402,733	402,733
Pseudo R^2	0.21	0.21	0.23	0.23

Probability of being chosen as the lead lender using the column (1) specification

		Past Lending Relationship	
	Average	No	Yes
Client Stock Holdings = 0	0.126	0.094	0.413
Client Stock Holdings = 1	0.158	0.120	0.479
Change in Probability	0.032	0.026	0.066

Table VIII: lead-lender probability

Table VIII—*Continued*

Probability of being chosen as the lead lender using the column (2) specification

		Past Lending Relationship	
	Average	No	Yes
Client Stock Holdings >1% = 0	0.135	0.101	0.441
Client Stock Holdings >1% = 1	0.180	0.136	0.525
Change in Probability	0.045	0.035	0.084

Columns (3) and (4) show that the results are robust to including bank (lender)-

Table VIII continued

5.9 Table IX: voting dissent and affiliated ownership

Read:

- Lender-affiliated fund ownership is negatively related to voting dissent on executive compensation proposals.
- The coefficient is around -0.545 in OLS and remains negative in tobit specifications.
- Nonlender-affiliated fund ownership is not similarly negative.

Economic message: Funds affiliated with lending banks appear to support borrower management, which can help preserve lending relationships.

Voting Dissent and Commercial Bank-Affiliated Funds Ownership

This table presents estimates of ordinary least squares (OLS) and tobit (with censoring at 0 and 1) firm-level panel regressions of voting dissent on executive compensation proposals. The dependent variable is the percentage of votes against management's proposals on executive compensation plans at shareholder meetings (*Voting Dissent*). *Lender-Affiliated Funds Ownership* is ownership by funds affiliated with commercial banks that were chosen by firm j as lead arrangers in a loan over the past three years. *Nonlender-Affiliated Funds Ownership* is ownership by funds affiliated with commercial banks that were not chosen by firm j as lead arrangers in a loan over the past three years. *Unaffiliated Funds Ownership* is ownership by funds unaffiliated with commercial banks. *Institutional ownership* is total institutional ownership and *Insider Ownership* is closely held shares. Ownership variables are defined as a percentage of market capitalization. All control variables are lagged by one period. Variable definitions are provided in Table A.I in the Appendix. The sample consists of non-U.S. firms for which votes at shareholder meetings are available in Institutional Shareholder Services/RiskMetrics (ISS) over the 2008 to 2010 period. Robust t-statistics adjusted for clustering at the country-industry level are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	OLS		Tobit	
	(1)	(2)	(3)	(4)
Lender-Affiliated Funds Ownership	-0.545^{**}	-0.520^{**}	-0.639^{**}	-0.642^{**}
	(−2.16)	(−2.12)	(−1.97)	(−2.04)
Nonlender-Affiliated Funds Ownership	−0.124		−0.128	
	(−1.28)		(−1.04)	
Unaffiliated Funds Ownership	0.092^{*}		0.107	
	(1.84)		(1.64)	
Institutional Ownership		0.043^{*}		0.065^{**}
		(1.75)		(2.05)
Insider Ownership	-0.023^{*}	−0.016	-0.032^{**}	−0.022
	(−1.71)	(−1.13)	(−2.09)	(−1.33)
log(Market Capitalization)	−0.003	−0.003	−0.001	−0.002
	(−1.18)	(−1.45)	(−0.35)	(−0.82)
Leverage	0.032^{*}	0.029	0.041^{*}	0.038^{*}
	(1.73)	(1.62)	(1.93)	(1.82)
Book-to-Market	0.002	0.002	0.001	0.000
	(0.33)	(0.29)	(0.10)	(0.05)
Return on Assets	-0.040^{**}	-0.040^{**}	-0.042^{**}	-0.041^{**}
	(−2.50)	(−2.42)	(−2.25)	(−2.25)
Year Fixed Effects	Yes	Yes	Yes	Yes
Stock Country Fixed Effects	Yes	Yes	Yes	Yes
Stock Industry Fixed Effects	Yes	Yes	Yes	Yes
Number of Observations	2,263	2,263	2,263	2,263
R^2	0.104	0.104		

Table IX: voting dissent and lender-affiliated ownership

5.10 Table X and Table XI: manager incentives and investor flows

Read:

- Table X studies fund manager turnover and conflict-related portfolio allocation.
- The results are consistent with the idea that managers who help the banking group face different career incentives.
- Table XI studies flows to commercial bank-affiliated funds and flow-performance sensitivity.
- Flows remain sensitive to performance, but the affiliated distribution channel can partially insulate funds from discipline.

Economic message: The organizational conflict can persist because managers and fund families face incentives that differ from maximizing fund-investor returns.

Commercial Bank-Affiliated Fund Manager Turnover and Portfolio Allocation to Client Stocks

This table presents estimates of fund-level probit regressions of fund manager turnover-performance sensitivity. The dependent variable is a dummy variable that takes a value of 1 if the fund manager is replaced in a quarter, and 0 otherwise (*Fund Manager Turnover*). *Commercial Bank-Affiliated* is a dummy variable that takes a value of 1 if the ultimate owner of the fund's management company is a commercial banking group, and 0 otherwise. *High Bias Fund* is a dummy variable that takes a value of 1 if an affiliated fund's *Bias in Client Stocks* is above the median in a given country and quarter, and 0 otherwise. *Bias in Client Stocks* is the portfolio bias in stocks of firms that borrow from the fund's parent bank versus the average weight of comparable passive funds. All control variables are lagged by one period. Variable definitions are provided in Table A.I in the Appendix. The sample consists of actively managed domestic equity mutual funds over the 2004 to 2010 period. Robust t-statistics adjusted for clustering at the ultimate owner level are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	All Funds (1)	Non-U.S. Funds (2)	U.S. Funds (3)
Commercial Bank-Affiliated	0.109***	0.149***	0.075
	(2.66)	(2.58)	(1.20)
High Bias Fund	−0.081	−0.188***	0.002
	(−1.41)	(−2.73)	(0.02)
Rank	−0.144***	−0.168**	−0.137**
	(−3.15)	(−2.29)	(−2.37)
Fund Manager Tenure	0.009***	0.024***	0.000
	(2.60)	(4.67)	(0.04)
log(TNA)	−0.039***	−0.024*	−0.048***
	(−3.67)	(−1.78)	(−3.72)
log(Family TNA)	0.058***	0.034*	0.064***
	(4.95)	(1.82)	(4.90)
log(1+Age)	−0.005	−0.059	0.032
	(−0.19)	(−1.53)	(1.15)
Flow	−0.003**	−0.002	−0.003**
	(−2.11)	(−0.95)	(−1.98)
Team-Managed	−0.140***	−0.220***	−0.106**
	(−4.12)	(−3.37)	(−2.55)
Year Fixed Effects	Yes	Yes	Yes
Country Fixed Effects	Yes	Yes	No
Number of Observations	72,373	26,052	46,321
Pseudo-R^2	0.055	0.102	0.035
Probability (fund manager left |fund survived) in quarter t			
High Bias Fund = 0	2.01%	2.75%	1.57%
High Bias Fund = 1	1.66%	1.82%	1.58%
Change in Probability	−0.35%	−0.93%	0.01%

Table X: manager turnover

Table XI

Flows to Commercial Bank-Affiliated Funds

esents estimates of ordinary least squares (OLS) regressions of fund flows (percentage growth in TNA). In columns (1), (3), and (5) nsists of those funds whose ultimate owner of the fund's management company is a commercial banking group (commercial bank ls). In columns (2), (4), and (6), the sample consists of all other funds (unaffiliated funds). The piecewise linear specification include ance rank segments: $Low = \min(0.2, Rank)$, $Mid = \min(0.6, Rank - Low)$, and $High = Rank - (Low + Mid)$. *Rank* is the fractiona rank ranging from 0 to 1, which is assigned according to the average four-factor alpha over the past four quarters in a given quarte All control variables are lagged by one period. Variable definitions are provided in Table A.I in the Appendix. The sample consists o ged domestic equity mutual funds over the 2000 to 2010 period. Robust t-statistics adjusted for clustering at the ultimate owner leve n parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	All Funds		Non-U.S. Funds		U.S. Funds	
	Affiliated (1)	Unaffiliated (2)	Affiliated (3)	Unaffiliated (4)	Affiliated (5)	Unaffiliatec (6)
	8.713***	7.107***	3.227	9.984***	12.149***	5.303**
	(4.31)	(3.93)	(1.38)	(3.91)	(3.91)	(2.34)
	3.929***	4.759***	2.847***	3.250***	4.793***	5.446***
	(8.39)	(12.02)	(4.58)	(5.32)	(7.88)	(10.68)
	10.632***	14.521***	14.541***	17.593***	5.427	13.010***
	(3.66)	(6.72)	(4.02)	(6.72)	(1.28)	(4.25)
	Yes	Yes	Yes	Yes	Yes	Yes
l Effects	Yes	Yes	Yes	Yes	Yes	Yes
l Effects	Yes	Yes	Yes	Yes	No	No
servations	41,046	78,378	21,474	25,979	19,572	52,399
	0.088	0.101	0.057	0.070	0.157	0.118

Table XI: flows to affiliated funds

6 Short summary

- The paper studies actively managed domestic equity mutual funds in 28 countries from 2000 to 2010.
- Commercial bank-affiliated funds are widespread internationally and especially important outside the United States.
- Bank-affiliated funds underperform unaffiliated funds by roughly 92 basis points per year.
- Underperformance is stronger when the parent bank's lending business is larger.
- Underperformance is also stronger when funds overweight the parent bank's lending clients.
- Divestitures of asset management divisions reduce client-stock holdings and improve performance.
- Client stocks bought by high-bias affiliated funds do not outperform; they underperform.
- Affiliated fund holdings increase the parent bank's chances of future lending business.
- Affiliated ownership reduces voting dissent against borrower management.
- Fund manager turnover and flow evidence support a career-concern and investor-discipline channel.
- The main lesson is that combining commercial banking and asset management creates conflicts that can harm fund investors.

7 Conclusion

7.1 Core conclusion

Ferreira, Matos, and Pires show that commercial bank-affiliated mutual funds underperform unaffiliated funds and that this underperformance is linked to affiliated funds' support of the parent bank's lending clients.

7.2 Economic intuition

A commercial banking group may benefit when its asset management division buys or holds the stocks of lending clients. This support can strengthen client relationships, improve future lending opportunities, and reduce borrower-management dissent. But these benefits accrue to the banking group, not necessarily to fund investors. Fund investors therefore bear the cost of cross-division conflicts of interest.

7.3 Takeaway

The paper provides international evidence that financial conglomerates can create hidden conflicts between asset management and commercial banking. The key result is not merely that affiliated funds perform poorly; it is that their poor performance is tied to observable portfolio support for the parent bank's borrowers.